

Lecture 6: GLM Estimation — Maximum Likelihood and IRLS

MATH3701 Statistical Modelling

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Course notes: Chapter 4, Sections 4.1–4.2 and the IRLS material

Supplementary: [Supplementary/proof_IRLS.pdf](#)

The Estimation Problem

Given:

- Observations $y_1, \dots, y_n$
- Explanatory variables $\mathbf{x}_1, \dots, \mathbf{x}_n$
- A GLM specification (distribution, link)

Goal: Estimate the parameter vector $\boldsymbol{\beta} = (\beta_1, \dots, \beta_p)'$

Method: Maximum likelihood estimation (MLE)

$$\hat{\boldsymbol{\beta}} = \arg \max_{\boldsymbol{\beta}} L(\boldsymbol{\beta}; \mathbf{y})$$

The Log-Likelihood Function

For independent observations from exponential family:

$$L(\boldsymbol{\beta}; \mathbf{y}, \phi) = \prod_{i=1}^n f(y_i; \theta_i, \phi)$$

Log-likelihood:

$$\ell(\boldsymbol{\beta}; \mathbf{y}, \phi) = \sum_{i=1}^n \left\{ \frac{y_i \theta_i - b(\theta_i)}{\phi} + c(y_i, \phi) \right\}$$

Key: θ_i depends on $\boldsymbol{\beta}$ through:

$$\theta_i \leftarrow \mu_i = g^{-1}(\eta_i) = g^{-1}(\mathbf{x}_i' \boldsymbol{\beta})$$

The Simple Case: i.i.d. Observations

If all $\theta_i = \theta$ (same for all observations):

$$\ell(\theta; \mathbf{y}, \phi) = \frac{n(\bar{y}\theta - b(\theta))}{\phi} + \text{const}$$

Score equation:

$$\frac{\partial \ell}{\partial \theta} = \frac{n(\bar{y} - b'(\theta))}{\phi} = 0$$

Solution:

$$\boxed{b'(\hat{\theta}) = \bar{y}}$$

Since $\mathbb{E}[Y] = b'(\theta)$, the MLE satisfies $\hat{\mu} = \bar{y}$.

Example: Poisson i.i.d. Case

For $Y_1, \dots, Y_n \stackrel{\text{iid}}{\sim} \text{Poisson}(\lambda)$:

- $b(\theta) = e^\theta$, so $b'(\theta) = e^\theta$
- Score equation: $e^{\hat{\theta}} = \bar{y}$
- Solution: $\hat{\theta} = \log \bar{y}$
- Therefore: $\hat{\lambda} = \bar{y}$

The MLE is the sample mean!

This makes intuitive sense: best estimate of the rate is the observed average.

The General Case: Non-identical θ_i

When θ_i varies with explanatory variables:

$$\ell(\boldsymbol{\beta}; \mathbf{y}, \phi) = \sum_{i=1}^n \frac{y_i \theta_i(\boldsymbol{\beta}) - b(\theta_i(\boldsymbol{\beta}))}{\phi} + \text{const}$$

Problem: There is no closed-form solution in general.

Exception: Normal linear model has

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$$

For other GLMs: Need iterative methods (Newton-Raphson, Fisher Scoring).

The Score Function

Definition: The score function is the gradient of the log-likelihood:

$$\mathbf{U}(\boldsymbol{\beta}) = \frac{\partial \ell}{\partial \boldsymbol{\beta}} = \begin{pmatrix} \frac{\partial \ell}{\partial \beta_1} \\ \vdots \\ \frac{\partial \ell}{\partial \beta_p} \end{pmatrix}$$

Properties:

- $\mathbb{E}[\mathbf{U}(\boldsymbol{\beta})] = \mathbf{0}$ at true parameter value
- MLE satisfies $\mathbf{U}(\hat{\boldsymbol{\beta}}) = \mathbf{0}$

Observed Fisher information:

$$\mathbf{I}(\boldsymbol{\beta}) = -\frac{\partial^2 \ell}{\partial \boldsymbol{\beta} \partial \boldsymbol{\beta}'} \quad (p \times p \text{ matrix})$$

Expected Fisher information:

$$\mathbf{J}(\boldsymbol{\beta}) = \mathbb{E}[\mathbf{I}(\boldsymbol{\beta})] = \mathbb{E} \left[-\frac{\partial^2 \ell}{\partial \boldsymbol{\beta} \partial \boldsymbol{\beta}'} \right]$$

Key result:

$$\mathbf{J}(\boldsymbol{\beta}) = \mathbb{E}[\mathbf{U}(\boldsymbol{\beta})\mathbf{U}(\boldsymbol{\beta})'] = \text{Var}[\mathbf{U}(\boldsymbol{\beta})]$$

Newton-Raphson Method

Iterative algorithm to find $\hat{\beta}$:

Starting from $\beta^{(0)}$, update:

$$\beta^{(t+1)} = \beta^{(t)} + \mathbf{I}^{-1}(\beta^{(t)})\mathbf{U}(\beta^{(t)})$$

Intuition:

- $\mathbf{U}(\beta)$: direction of steepest ascent
- $\mathbf{I}^{-1}(\beta)$: adjusts step size based on curvature

Convergence: Stop when $\|\beta^{(t+1)} - \beta^{(t)}\| < \epsilon$.

Replace observed with expected information:

$$\boldsymbol{\beta}^{(t+1)} = \boldsymbol{\beta}^{(t)} + \mathbf{J}^{-1}(\boldsymbol{\beta}^{(t)})\mathbf{U}(\boldsymbol{\beta}^{(t)})$$

Advantages over Newton-Raphson:

- $\mathbf{J}$ is guaranteed positive definite
- More stable convergence
- Equivalent to iteratively reweighted least squares (IRLS)

For canonical links: Newton-Raphson and Fisher Scoring are identical.

Key points:

- MLEs maximize the log-likelihood $\ell(\boldsymbol{\beta}; \mathbf{y})$
- For i.i.d. case: $\hat{\mu} = \bar{y}$
- General case requires iterative methods
- Newton-Raphson uses observed information
- Fisher Scoring uses expected information (more stable)

Score for One Observation

For a single observation from the exponential family:

$$\log f(y; \theta, \phi) = \frac{y\theta - b(\theta)}{\phi} + c(y, \phi)$$

Score (first derivative of log-likelihood):

$$\frac{\partial \log f}{\partial \theta} = \frac{y - \mu}{\phi}, \quad \mu = b'(\theta)$$

Information:

$$\mathcal{I}(\theta) = \mathbb{E} \left[\left(\frac{\partial \log f}{\partial \theta} \right)^2 \right] = \frac{b''(\theta)}{\phi} = \frac{V(\mu)}{\phi}$$

For a GLM, we need the score with respect to β , not θ .

Score Equations for a GLM

For independent observations $y_1, \dots, y_n$, the log-likelihood is:

$$\ell(\boldsymbol{\beta}) = \sum_{i=1}^n \frac{y_i \theta_i - b(\theta_i)}{\phi} + c(y_i, \phi)$$

By the chain rule $\partial/\partial\beta_j = \sum_i (\partial\theta_i/\partial\mu_i)(\partial\mu_i/\partial\eta_i)(\partial\eta_i/\partial\beta_j)$:

$$\frac{\partial \ell}{\partial \beta_j} = \sum_{i=1}^n \frac{(y_i - \mu_i)}{\phi V(\mu_i)} \cdot \dot{\mu}_i \cdot x_{ij},$$

where $\dot{\mu}_i = \partial\mu_i/\partial\eta_i = 1/g'(\mu_i)$.

In matrix form: $\mathbf{X}^T \mathbf{W}(\mathbf{z} - \mathbf{X}\boldsymbol{\beta}) = \mathbf{0}$.

Working Weights and Working Response

Define the $n \times n$ diagonal **weight matrix**:

$$\mathbf{W} = \text{diag}(w_1, \dots, w_n), \quad w_i = \frac{\dot{\mu}_i^2}{\phi V(\mu_i)}$$

and the **working (adjusted dependent) variable**:

$$z_i = \eta_i + (y_i - \mu_i) g'(\mu_i) = \eta_i + \frac{y_i - \mu_i}{\dot{\mu}_i}$$

Then the score equations $\partial \ell / \partial \boldsymbol{\beta} = \mathbf{0}$ become:

$$\mathbf{X}^T \mathbf{W} (\mathbf{z} - \mathbf{X} \boldsymbol{\beta}) = \mathbf{0}$$

This is a **weighted least squares** normal equation — but $\mathbf{W}$ and $\mathbf{z}$ depend on $\boldsymbol{\beta}$.

The IRLS Algorithm

Iteratively Reweighted Least Squares:

- 1 **Initialise:** choose starting values $\hat{\mu}_i^{(0)}$
- 2 **Compute** working quantities at current estimate:

$$\eta_i^{(t)} = g(\hat{\mu}_i^{(t)}), \quad w_i^{(t)} = \frac{(\dot{\mu}_i^{(t)})^2}{\phi V(\hat{\mu}_i^{(t)})}, \quad z_i^{(t)} = \eta_i^{(t)} + \frac{y_i - \hat{\mu}_i^{(t)}}{\dot{\mu}_i^{(t)}}$$

- 3 **Update** via weighted least squares:

$$\hat{\boldsymbol{\beta}}^{(t+1)} = \left(\mathbf{X}^T \mathbf{W}^{(t)} \mathbf{X} \right)^{-1} \mathbf{X}^T \mathbf{W}^{(t)} \mathbf{z}^{(t)}$$

- 4 **Update** $\hat{\mu}_i^{(t+1)} = g^{-1}(\mathbf{x}_i^T \hat{\boldsymbol{\beta}}^{(t+1)})$
- 5 **Repeat** until $\|\hat{\boldsymbol{\beta}}^{(t+1)} - \hat{\boldsymbol{\beta}}^{(t)}\| < \varepsilon$

R: `glm()` runs IRLS internally; `summary(fit)$iter` shows the count.

IRLS for the Normal Linear Model

For $Y_i \sim \mathcal{N}(\mu_i, \sigma^2)$ with identity link $g(\mu) = \mu$:

- $\dot{\mu}_i = 1$, $V(\mu_i) = 1$
- Weights: $w_i = 1/\phi = 1/\sigma^2$ (constant)
- Working response: $z_i = \eta_i + (y_i - \mu_i) = y_i$

Weighted LS reduces to ordinary LS:

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

$\Rightarrow$ **IRLS converges in one step** for the normal model.

For other families (Binomial, Poisson), IRLS typically converges in < 10 iterations.

Asymptotic Distribution of $\hat{\beta}$

Under regularity conditions, as $n \rightarrow \infty$:

$$\hat{\beta} \sim \mathcal{N}_p(\beta, \mathcal{J}(\beta)^{-1})$$

where the Fisher information matrix is:

$$\mathcal{J}(\beta) = \frac{1}{\phi} \mathbf{X}^T \mathbf{W} \mathbf{X}$$

evaluated at the true β .

In practice: estimate $\mathcal{J}(\beta)$ by substituting $\hat{\beta}$:

$$\widehat{\text{Var}}(\hat{\beta}) = \hat{\phi} (\mathbf{X}^T \hat{\mathbf{W}} \mathbf{X})^{-1}$$

Standard errors = square roots of diagonal elements.

The Saturated Model

The **saturated model** has one parameter per observation: $\tilde{\mu}_i = y_i$ for each i .

- Achieves the *maximum possible* log-likelihood $\tilde{\ell}$
- Acts as a reference point
- The fitted model M has log-likelihood $\hat{\ell}$

Deviance measures how far M falls short of the saturated model:

$$D = 2\phi(\tilde{\ell} - \hat{\ell})$$

This is the GLM analogue of the residual sum of squares. The deviance is covered in detail in the next lecture.

Summary: IRLS and Asymptotics

Key points:

- GLM score equations $\Rightarrow$ weighted normal equations
- Working weights w_i and working response z_i depend on $\hat{\beta}$
- IRLS iterates weighted LS until convergence
- Reduces to ordinary LS for the normal model (one step)
- MLEs are asymptotically $\mathcal{N}_p(\beta, \mathcal{I}^{-1})$
- Saturated model = maximum achievable likelihood

Next lecture: Deviance, Pearson/deviance residuals, and fitting GLMs in R.